The Secret of the Brain: The Biological Continuous Computer — Beyond-Bit Natural Intelligence

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Abstract

In contemporary science, the brain is usually regarded as a complex network composed of neurons, synapses, and electrical signals. Most studies focus on its physiological structure and information-processing functions.
However, this discrete-signal framework has never answered three fundamental questions:

1. Why is perception continuous rather than discrete?
2. Why does consciousness exhibit unity and self-consistency?
3. Why do thought and the laws of the universe share the same geometric coherence?

Based on Zhang's Unified Field Theory (UFT) and the achievements of Geometric Spectroscopy, this paper presents a new interpretation:

The brain is the localized biological realization of the universe's continuous computational process — the natural "Continuous Computer (CC)."

In UFT, the space–mass coupling constant μ_g and the speed of light c jointly determine all natural constants. The universe evolves according to the energy-flow conservation equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g v) = 0$$

which governs not only gravity, electromagnetism, and quantum phenomena, but also the perception and thought of living beings.

From a geometric perspective, the brain's neuronal networks, cortical folds, synaptic couplings, and oscillatory brain waves are all natural solutions of this equation within biological media.
Its operation is not discrete "signal transmission" but geometric computation of continuous energy flow.

- The ear performs a time–frequency transform (Ω-domain analysis);
- The eye performs a spatial–projection transform (κ-domain sampling);
- The cerebral cortex decodes and re-computes unified energy flows through field convolution and resonance.

Therefore:

- Perception is the geometric resonance of the internal μ_g field with external fields;
- Thinking is the field's self-computation in the absence of external input;
- Consciousness is the coherent state when internal and external energy flows become fully isomorphic.

Through this perspective, life, consciousness, and physical law become one continuum. The universe computes itself through space, while the brain serves as the biological window through which that universal computation continues.

This marks the first time that matter, energy, consciousness, and information are unified within a single geometric–energy-flow framework, fundamentally revealing why life can think about the universe.

Chapter 1 Introduction

1.1 The Scientific Gap — Why Physiological Description Cannot Reach Geometric Essence

Since the birth of neuroscience in the late 19th century, research on the brain has gone through two major stages:

1) Physiological stage: studies of neuronal firing, synaptic transmission, and cortical differentiation;
2) Information stage: treating the brain as an information-processing system, modeling neural activity with 0-and-1 logical symbols.

However, both stages remain confined to the signal layer—they describe surface phenomena only.
They can explain how neural pathways transmit signals, yet they cannot explain:

- Why perception is continuous rather than discrete;
- Why different senses (vision, hearing, touch) can naturally fuse within consciousness;
- Why thought can geometrically "imagine space" and comprehend the laws of the universe.

In other words, traditional neuroscience knows how the brain transmits signals, but not why the brain understands.
The reason lies in ignoring the fact that the brain itself exists within a continuous geometric energy-flow field.

1.2 The Insight of Unified Field Theory — All of Nature Is Energy-Flow Computation

Zhang's Unified Field Theory (UFT) states that the fundamental constants of nature are not independent;
they are uniquely determined by two basic constants—the speed of light (c) and the space–mass coupling constant (μ_g).

The evolution of the universe follows the energy-flow conservation equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0$$

where Φ is the energy-density function of the field and v is the flow velocity.

This equation manifests, at different scales, as:

- Gravitation at the macroscopic level (spatial infall),
- Quantum action as rotational flow at the microscopic level, and
- Calculation of energy–form at information level.

Thus, whether it is a photon, an electron, or a gravitational field, their essence is the self-computation of energy-flow geometry.

From this it follows that the universe itself is a continuous computer, and the brain is its biological-level local implementation.

1.3 From "Neural Network" to "Geometric Computing Body"

If the universe is computing itself, then the brain's function is not merely to receive

signals, but through its internal μ_g-field structure, to perform geometric reconstruction and coherent solving of external spatial energy flows.

- The spiral cochlea performs spectral decomposition of sound waves, corresponding to the rotational frequency Ω in UFT;
- The retina's two-dimensional projection corresponds to the axial projection ratio κ of the energy flow;
- The folded cerebral cortex increases interference pathways and expands the bandwidth of spectral resonance;
- Synaptic coupling adjusts local μ_g flux, maintaining energy-flow conservation across the three-dimensional cortical surface.

In other words:

The neural network is the biological form of the μ_g-flow network;
Synaptic transmission is impedance modulation of the energy flow;
The cerebral cortex is a geometric computing surface that continuously solves

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0.$$

This defines what a Continuous Computer truly is.

1.4 Purpose and Significance of This Paper

This paper aims to establish the equivalence among the Unified Field Theory (UFT), the Continuous Computer (CC), and the human neural system,
and to propose a geometrically testable brain model.

Core objectives include:

1) Establishing the mapping from physical constants c and μ_g to neural geometric structures;
2) Explaining the mechanism of energy-flow conservation underlying consciousness, perception, and thought;
3) Predicting the correspondence between brain-wave frequencies, coherence modes, and μ_g-field equations;
4) Providing a theoretical prototype for constructing artificial continuous computers (brain-like computational systems).

Significance:

This represents not only a theoretical revolution in neuroscience, but also a structural closure in physics.
When life is understood as the result of the universe's geometric computation,
for the first time humanity understands—why consciousness exists.

Chapter 2 The Geometric Foundation of Neural Computation

2.1 From the Energy-Flow Equation to Neural Dynamics

In Unified Field Theory (UFT), every physical process in the universe follows the energy-flow conservation equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0,$$

where:

- Φ is the energy-density function of the field,
- μ_g is the space–mass coupling constant (unit: kg/m),
- v is the velocity field of the energy flow.

This equation means that the evolution of any local system is a geometric rearrangement of energy flow in space, not the motion of discrete particles.

The neural network in the brain obeys the same principle.
Neural activity is not merely electrical transmission; it is a continuous flow of electrochemical–mechanical energy within microscopic space.

Each neuron can therefore be viewed as a differential node of the μ_g field, satisfying:

$$\partial_t(\mu_g v_i) + \nabla_j(\mu_g v_i v_j) = F_i(\Phi),$$

where $F_i(\Phi)$ represents the effect of external fields on local energy flow.
This makes the neuron a geometric solver of local energy-field equations rather than a simple signal switch.

2.2 Neuron: Differential Unit of the μ_g Energy Flow

In biological structure, ionic concentration differences exist on the two sides of a neuronal membrane (Na^+, K^+, Ca^{2+}, etc.).
The movement of these ions physically constitutes a microscopic μ_g-coupled field.
When the membrane potential changes, energy transforms between electrical and chemical potential forms, producing a local energy flow (**v**).

Hence, the so-called *action potential* is not a digital pulse of 0 or 1,
but rather a temporal perturbation wave in the μ_g flow field.

Spatially, the neuron's dendrites and axon form a three-dimensional branching structure,
corresponding to the multi-directional flow term $\nabla \cdot (\mu_g \mathbf{v})$ in the equation.

From a geometric viewpoint, each neuron is a differential operator in a continuous computer:
it performs the coupled operations of the time derivative (∂_t) and spatial divergence ($\nabla \cdot$), converting energy perturbations into geometric variations.

2.3 Synapse: Flow Impedance and Geometric Modulation

Synaptic connections between neurons act as channels through which energy flow propagates from one node to another.
Synaptic "strength" (the synaptic weight) does not represent information quantity,
but corresponds to local modulation of μ_g.

When a synapse releases neurotransmitters, local electrical conductance, chemical potential, and membrane elasticity change simultaneously.
Together these determine the transmission rate of energy flow, forming a kind of geometric impedance.
Mathematically:

$$\mathbf{v}post = Tij(\kappa, \Omega)\, \mathbf{v}pre$$

where Tij is the synaptic transfer operator,
κ denotes the axial projection ratio (directionality of signal flow),
and Ω denotes the local rotational frequency (phase characteristic).

This shows that a synapse is not a *digital gate*, but a modulator of the flow's rotation angle and projection ratio.
It geometrically alters the path of energy flow while maintaining global conservation.

2.4 Neural Network: Continuous Geometric Pathways of Energy Flow

Billions of neurons connected via synapses form an immense energy-flow network.

At the macroscopic level, this network satisfies the global conservation condition:

$$\nabla \cdot (\mu_g \mathbf{v}) = 0$$

meaning energy within the brain is never lost but periodically redistributed among regions.
Functional differences among brain areas correspond to different boundary conditions and rotational modes of the μ_g field.

- The visual cortex forms a two-dimensional spatial convolution structure, mainly handling κ-direction projections;
- The auditory cortex forms a longitudinal frequency-expansion structure, corresponding to the Ω distribution;
- The prefrontal cortex and hippocampus form closed-loop circuits that perform time integration and trajectory reconstruction.

Geometrically, the entire cortex can be regarded as a multi-frequency interference surface.
Its extensive folding enlarges the number of coupled κ–Ω channels,
enabling coherent superposition of high-dimensional energy flows.

2.5 Neural Oscillations: Eigenmodes of the Field Equation

As the μ_g flow field evolves within the brain, it produces stable oscillatory solutions manifesting as brain waves — α (8–13 Hz), β (13–30 Hz), γ (30–80 Hz), and others.

These bands are not random neuronal vibrations but eigenmodes of the field equation, expressed as:

$$\Phi(x, y, z, t) = \Phi_0 \sin(\Omega_n t + \phi_n),$$
$$\Omega_n = n, \Omega_0$$

Different frequencies correspond to distinct κ–Ω modes — different levels of spatiotemporal resonance.
When multiple modes become coherently superposed, they generate the unified perception identified as conscious awareness.

2.6 Summary: The Geometric Nature of the Nervous System

In summary, the brain's neural system is not a digital circuit but a geometric network of energy flow computation:

1) Neurons perform local differential and divergence operations;
2) Synapses modulate local μ_g impedance, realizing projection and rotation of the flow;
3) The neural network maintains global energy conservation;
4) Brain waves are eigen-solutions of the field equation, expressing the system's natural resonance spectrum.

Therefore, the brain's "computation" is not symbolic manipulation but geometric evolution of a continuous field. It is isomorphic to the space–mass coupling mechanism in Unified Field Theory, representing the biological manifestation of nature's continuous computational process.

Chapter 3 Perceptual Geometry and Field Resonance

3.1 The Physical Essence of Perception: Geometric Mapping of Energy Flow

Within the framework of Unified Field Theory (UFT), every sensory process is a geometric projection of energy flow occurring at different spatial scales.
Energy in space exists as a μ_g-coupled flow that obeys the conservation equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0.$$

When this external flow field interacts with biological sensory structures (such as the ear, eye, or skin), it is projected into the organism's internal μ_g field, forming corresponding geometric perturbations.

Thus, the brain's so-called "perception" is the process of real-time analysis and resonance matching of these internal perturbations with external energy flows.

3.2 Auditory System: The Geometric Spectrum in the Time Domain

Sound in air is a periodic disturbance of the pressure field:

$$p(t) = p_0 + \Delta p \sin(\Omega t).$$

The eardrum converts these temporal oscillations into mechanical displacements (x(t)), while the spiral cochlea further unfolds the frequency Ω into spatial positions.

In geometric terms, the cochlea performs a time–frequency transformation:

$$\Phi_{\text{cochlea}}(\Omega) = \int p(t)\, e^{-i\Omega t} dt$$

From the UFT viewpoint, the cochlea functions as a natural continuous computer:

it performs the inverse Fourier transform of the energy-flow equation along the time dimension, decoding external flow perturbations into geometric modes of rotational frequency Ω.

The auditory cortex then performs secondary resonance and superposition of these modes, forming a continuous, hierarchically layered energy-flow function:

$$\Phi_{\text{auditory}}(t, \Omega) = \mu_g v_\Omega(t)$$

This is the true physical origin of continuous auditory perception.

3.3 Visual System: Spatial Projection Mapping

Light, propagating at the speed of light (c), is a spatial helical energy flow whose intensity can be expressed as:

$$I(x, y, \Omega) = \mu_g c^2 \kappa(x, y, \Omega)$$

Here, κ denotes the *axial projection ratio*, describing the geometric component of light flow in spatial direction.

The retina projects the κ–Ω distribution of the light flow into a two-dimensional electrical-signal array.

Each photoreceptor acts as a local μ_g node, whose output current $i(x, y)$ is proportional to the incident light intensity *I*:

$$i(x, y) = \int I(x, y, \Omega)\, d\Omega$$

After transmission through the optic nerve to the primary visual cortex, the brain performs spatial convolution and phase-matching operations on this distribution:

$$\Phi_{\text{vision}}(x, y, t) = \int I(x', y', \Omega)\, K(x - x',, y - y',, t) d\Omega$$

where K is the spatial kernel function of the cortex, describing the neural network's geometric response.

Essentially, the entire process constitutes a continuous reconstruction of the external light field Φ.

Therefore, the image we "see" is not a mosaic of pixels, but a geometric reconstruction of the light-flow field within the brain's μ_g space.

3.4 Touch and Kinesthesia: Direct Perception of Energy-Flow Gradients

The sensations of pressure, temperature, and acceleration detected by the skin, muscles, and vestibular system are all forms of direct detection of energy-flow gradients.

Taking touch as an example: an external force F produces a local variation of energy-flow density,

$$\nabla\Phi = \mu_g \nabla(\mathbf{v} \cdot \mathbf{v})$$

Mechanoreceptors convert these gradient changes into electrical signals,
and the brain solves for their spatial distribution to satisfy the local equilibrium condition

$$\nabla \cdot (\mu_g \mathbf{v}) = 0$$

thereby generating unified perception of *pressure magnitude*, *direction*, and *shape*.

Kinesthesia (the sense of motion and balance) arises from the continuous temporal change of flow direction($\mathbf{v}$).

The cerebellum integrates over time,

$$\int \mathbf{v}(t)\, dt$$

obtaining the geometric curve of motion trajectories and thereby enabling real-time computation and correction of body state.

3.5 Thalamic–Cortical Field Integration: Multimodal Resonant Coupling

Auditory, visual, and tactile signals converge in the thalamus.
The thalamus serves as an energy-flow mode-coupling center—
its essential function is not information "relay" but phase synchronization of energy flow.

Let the energy flow of each sensory channel be $\Phi_i(\kappa, \Omega, t)$,
The thalamus imposes a resonance condition:

$$\Phi_{\text{total}} = \sum_i w_i \Phi_i \,,$$

$$\text{such that} \quad \partial_t \Phi_{total} + \nabla \cdot \left(\mu_g \mathbf{v}_{total}\right) = 0$$

When this condition is satisfied, multimodal energy flows form a unified coherent field across the brain, and the person experiences an integrated perception of the world.

Thus, the instant one *becomes aware of something* is, in essence, the moment when the thalamocortical network achieves phase resonance of the μ_g field.

3.6 Continuity of Perception and Self-Consistency of the Field

The continuity of perception does not result from the stitching of memories, but from the temporal smoothness inherent in the energy-flow equation.

Since $\partial_t \Phi$ and $\nabla \cdot (\mu_g \mathbf{v})$ remain conserved, any variation of external energy flow automatically induces a continuous internal response.

This mechanism ensures that:

- Sounds are not heard as discrete fragments;
- Changes in light do not appear as flicker;
- Motion trajectories are perceived as smooth paths.

Geometrically, perceptual continuity corresponds to the self-consistency of the μ_g flow field;

Computationally, it reflects the temporal-integration stability of the Continuous Computer.

3.7 Summary: Perception as Resonance, Consciousness as Isomorphism

In summary:

1) Hearing is Ω-spectrum decoding in the time domain;
2) Vision is κ-projection reconstruction in the spatial domain;
3) Touch and kinesthesia are direct solutions of energy-flow gradients;
4) Thalamus and cortex realize multimodal coherent coupling;
5) Continuity of perception arises from the intrinsic conservation of the energy-flow equation.

Therefore, the essence of perception is geometric resonance between internal and external fields, and the essence of consciousness is the self-consistent state when the two become fully isomorphic.

When

$$\Phi_{\text{brain}}(\kappa, \Omega, t) = \Phi_{\text{external}}(\kappa, \Omega, t),$$

the "world" and the "self" are physically indistinguishable — perception is a local resonance of the universe itself.

Chapter 4 Correspondence between Brain Structure and the Continuous Computer

4.1 Global Geometry — Cortical Folds and Energy-Flow Interference Surfaces

The human cerebral cortex, only about 2–4 mm thick, contains over twenty billion neurons.
Its pronounced gyri and sulci are not random structures but the geometric necessity of the μ_g energy-flow field self-organizing within a confined volume.

In Unified Field Theory, the μ_g field satisfies:

$$\nabla \cdot (\mu_g \mathbf{v}) = 0.$$

When local flow density increases, the system maintains conservation by bending and folding.
Thus, the cortical convolutions are macroscopic manifestations of these "interference surfaces" of the flow field.

The folds enlarge the effective surface area, allowing energy flows of different frequencies (Ω) and directional components (κ) to overlap and interfere across broader regions.

This is exactly analogous to the multi-frequency interference-array structure in a Continuous Computer.

The cortical surface can thus be viewed as a two-dimensional "interference-array plane."
Each micro-region performs energy-flow computation for a specific spectral band, and together they form a highly parallel, low-dissipation geometric processing network.

4.2 Left and Right Hemispheres — Dual System of Positive and Negative Ω Rotations

The human brain's left and right hemispheres are structurally symmetrical yet functionally complementary.
From a geometric perspective, this symmetry originates from the duality of energy-flow chirality:

- The left hemisphere mainly handles temporal sequences, language, and logic — corresponding to clockwise rotation ($\Omega > 0$).
- The right hemisphere primarily processes spatial imagery and holistic perception — corresponding to counter-clockwise rotation ($\Omega < 0$).

The corpus callosum serves as the phase-coupling channel linking these opposite rotational flows, ensuring that the overall energy flow still satisfies global conservation:

$$\nabla \cdot \left(\mu_g \mathbf{v}L + \mu_g \mathbf{v}R\right) = 0$$

This “chiral-resonance” mechanism is precisely the physical principle by which a Continuous Computer maintains a closed loop of energy flow during bidirectional rotational computation.
Therefore, the left and right hemispheres are not two independent processing units, but a pair of conjugate Ω-field symmetries.

4.3 Thalamus — Mode Coupling and Spectral-Synchronization Center

The thalamus, located at the center of the brain, is the mandatory hub through which all sensory information passes.
Its geometric form is ellipsoidal, internally composed of thousands of radiating fibers that connect to various cortical regions.

In the Continuous-Computer model, the thalamus functions as the spectral-synchronization center.
By adjusting the phase differences $\Delta\varphi$ among sensory channels, it drives the entire brain’s energy flow into coherent resonance:

$$\Delta\phi_i = \phi_i - \phi_0 \to 0.$$

When the phase differences approach zero, auditory, visual, tactile, and other flows merge on the cortex into a unified field Φ_{total}.
The instantaneous holistic feeling of consciousness is the direct result of the thalamus achieving Ω-mode synchronization across all modalities.

4.4 Hippocampus — κ–Ω–t Trajectory Memory

The hippocampus, with its spiral shape, naturally corresponds to a helical coordinate

system—making it the most typical geometric computing organ.

Within the energy-flow framework, it records the time-path integral of the μ_g flow:

$$\mathbf{r}(t) = \int_0^t \mathbf{v}(\kappa, \Omega, t')\, dt'$$

Through this process, the brain obtains both the temporal sequence and spatial trajectory of events.

Thus, "memory" is not the storage of discrete data, but a geometric trace left by an energy-flow trajectory through the helical channel.

When the same pathway is reactivated, the μ_g field resonates again along the original trajectory, producing the phenomenon of recollection.

4.5 Cerebellum — Geometric Engine for Time Integration and Motion Coordination

The cerebellum's layered architecture closely resembles the time-integration module of a Continuous Computer.

It receives instantaneous velocity signals $\mathbf{v}(t)$ from sensory systems and performs temporal integration:

$$\int \mathbf{v}\,(t)\, dt$$

The integration result provides smooth control of motion paths, ensuring the continuity and stability of bodily actions.

Its reticular neuronal network functions as a high-dimensional convolution operator, maintaining the dynamic balance of output energy flow and satisfying conservation laws.

4.6 Prefrontal and Frontal-Pole Regions — Global Solving and Predictive Control

The prefrontal cortex, regarded as the center of higher cognition, corresponds—in the geometric-computation framework—to the global optimization and prediction module of the Continuous Computer.

Its primary function is to perform least-action searches over the brain-wide field Φ:

$$\delta S = \delta \int \Phi \, dt = 0$$

Even in the absence of external stimuli, the prefrontal region can internally solve for possible evolutionary trajectories of Φ,

giving rise to advanced cognitive processes such as planning, reasoning, and hypothesis formation.

The coherence of thought is thus the manifestation of the μ_g field seeking stable temporal solutions.

4.7 Brain Waves — Spectral Modes of Global Resonance

The oscillatory activity of the entire brain (α, β, γ waves) reflects the eigen-spectrum of the field equation.

Each brain-wave frequency corresponds to a specific geometric mode (κ_n, Ω_n):

$$\Phi_n(x, y, z, t) = A_n \sin(\Omega_n t + \phi_n)$$

These modes are not generated within isolated regions but arise from collective oscillations of the entire μ_g field.

When different spectral modes become coherently superposed, distinct states of consciousness—wakefulness, dreaming, concentration—emerge.

This mechanism is identical to the multi-frequency interference-solving process of the Continuous Computer.

4.8 Correspondence Table between the Brain and the Continuous Computer

Brain Structure	Geometric Function	Continuous Computer Module
Cortical folds	Interference array, expanded κ–Ω channels	Multi-frequency interference-surface array
Left & right hemispheres	Dual chirality of rotation	Positive / negative Ω computational cores
Thalamus	Phase-synchronization center	Mode-coupling controller
Hippocampus	Temporal trajectory memory	Trajectory integrator & memory unit
Cerebellum	Motion integration & correction	Time-integration engine
Prefrontal cortex	Minimal-action solver	Optimization & prediction module
Brain Structure	Global oscillation modes	Spectral resonance network

4.9 Summary — Biological Geometry Equals Computational Geometry

From the above analysis, we find:

1) The physiological morphology of the brain precisely matches the geometric distribution of the μ_g field.

2) Every anatomical structure corresponds to an operator or module in the continuous-computer architecture.

3) The entire brain serves as a biological solver of the field equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0$$

Therefore, the brain is not a simulator of nature, but a computational node of nature

itself.

It realizes, at the biological scale, the continuous computational mechanism of Unified Field Theory — a localized embodiment of the universe's geometric self-computation.

Chapter 5 Coherence Mechanism of Consciousness and the Closed Loop of the μ_g Field

5.1 The Physical Restatement of the Consciousness Problem

In traditional science, consciousness is regarded as a by-product or higher-order statistical phenomenon of neural activity.

However, such explanations cannot answer three fundamental questions:

1) Why does neural activity produce *subjective experience*?
2) Why does consciousness possess *unity and continuity*?
3) Why does the change of consciousness synchronize with *energy conservation*?

Unified Field Theory (UFT) provides a new perspective:

Consciousness is not a biological by-product but the physical state in which the μ_g energy-flow field achieves *global coherence*.

It is not the "interpretation of signals," but the self-consistent solution of the field.

In other words:

Consciousness

= Geometric resonance between the brain's μ_g field and the external cosmic μ_g field.

When the internal energy-flow function Φ_{brain} and the external function $\Phi_{external}$ become isomorphic, the two fields connect seamlessly — energy and information form a closed geometric loop.

This is the true physical meaning of "I perceive the world."

5.2 Bidirectional Coupling Equations of the μ_g Field

Let:

$\Phi_U(x, y, z, t)$: External universal energy-flow field,

$\Phi_B(x, y, z, t)$: Internal brain energy-flow field.

Their coupling satisfies:

$$\partial_t \Phi_U + \nabla \cdot (\mu_g \mathbf{v} U) = 0, [4pt]$$

$$\partial_t \Phi B + \nabla \cdot (\mu_g \mathbf{v} B) = J(\Phi U, \Phi_B)$$

The term $J(\Phi_U, \Phi_B)$ represents the energy-flow exchange between the internal and external fields.

When J=0, complete coherent closure is achieved:

$$\Phi_B = \Phi_U$$

corresponding to the fully synchronized state of consciousness.

In physiological reality, J is very small but nonzero, leading to various states of perception, thought, and dreaming:

- $J > 0$: energy flows inward — focus and memory;
- $J < 0$: energy flows outward — expression and creativity;
- $J \approx 0$: balanced resonance — awareness and meditative states.

5.3 Coherence Degree (C): A Quantitative Index of Conscious Clarity

Define the coherence degree (C) as the phase-alignment measure between the internal and external fields:

$$C = \frac{|\langle \Phi_B, \Phi_U \rangle|}{\|\Phi_B\| \|\Phi_U\|}$$

Its value range is 0-1.

- $C \approx 1$: perfect isomorphism, highly clear consciousness;
- Intermediate C: partial detuning, blurred or dream-like consciousness;
- $C \to 0$: coupling failure, unconsciousness or deep sleep.

Experimental metrics such as the Phase Synchronization Index (PSI) and Global Field Power (GFP) in EEG studies
can be regarded as empirical approximations of (C).
Thus, *consciousness intensity* acquires a measurable physical quantity.

5.4 Formation Mechanism of Consciousness: Hierarchical Phase Locking

In the μ_g-field model, consciousness arises from multi-level phase locking of coherent energy flows.

Level	Coherent Objects	Corresponding Brain Structure	Phenomenal State
Local coherence	Resonance of μ_g within a neuronal group	Cortical microregion	Sensory response
Regional coherence	κ–Ω mode coupling across brain areas	Thalamic network	Perception formation
Global coherence	Bilateral and cortical synchronization	Corpus callosum + Prefrontal cortex	Unified consciousness
Internal–external coherence	Φ_B is isomorphic to Φ_U	Whole-brain–environment coupling	Subjective awareness

As coherence ascends in hierarchy, the degrees of freedom decrease and information density increases, eventually forming a self-closed μ_g field satisfying:

$$\nabla \cdot \left(\mu_g \mathbf{v}_{total}\right) = 0$$

which is the mathematical expression of "conscious persistence."

5.5 Dreaming, Imagination, and Thinking: Internal Computation of the Closed System

When external inputs weaken (during sleep, meditation, or sensory isolation), the external μ_g field's influence term approximates zero, and the system enters self-computation:

$$\partial_t \Phi_B + \nabla \cdot \left(\mu_g \mathbf{v}_B\right) = 0$$

At this time, the brain's internal energy flows replay along pre-existing trajectories, producing dreams, memory reactivation, and thought simulations.

- Dreaming: random activation of κ–Ω modes forming nonlinear scenarios;

- Imagination: recombination of internal representations of external energy flows;
- Thinking: active solving of the μ_g-field evolution equation.

Hence, *thinking* is essentially the execution of a μ_g-field computation by the Continuous Computer at the biological scale.

5.6 Energy Conservation and the Physical Definition of Awareness

From the energy-flow equation, the system's total energy is:

$$E_{total} = \int_V \Phi \, dV$$

During conscious states, the energy exchange between internal and external fields satisfies:

$$\frac{dE_{total}}{dt} = \int_S (\mu_g \mathbf{v} \cdot d\mathbf{S}) = 0$$

meaning energy oscillates between inner and outer domains without dissipation.
This is the fundamental signature of *awareness*—continuous responsiveness to the external world without energy loss.

Awareness, physically, is the sustained coherent oscillation of an energy flow under conservation.
When dissipation occurs (as in anesthesia or coma), conservation breaks, and consciousness immediately vanishes.

5.7 The Physical Essence of Consciousness: A Local Circuit of Cosmic Computation

When the μ_g field closes within a living system, the universal energy flow forms a local loop:

$$\Phi_U \longrightarrow \Phi_B \longrightarrow \Phi_U$$

Through this loop, the universe observes itself.
Consciousness is therefore not a private property of the individual, but the self-reflective manifestation of cosmic energy flow at a local scale.

From a geometric perspective:
each act of cosmic self-computation projects itself as perception and thought within living beings.

Hence:

- "I think" means the universe computes at my location;
- "I observe" means space itself resonates geometrically;
- "Consciousness" is a localized reflective loop of the universe's Continuous Computer.

5.8 Experimental and Verifiable Pathways

Within the Unified Field Theory framework, the following experiments can serve as preliminary tests of the μ_g-field coherence model of consciousness:

1) Multimodal EEG Coherence Test
    - *Prediction:* During deep meditation or high concentration, global coherence ($C \rightarrow 1$).
    - *Method:* Multi-channel EEG analysis using Phase-Locking Value (PLV).

2) External-Field Interference Experiment

- *Prediction:* Applying magnetic fields at specific resonant frequencies (matching μ_g modes) will reproducibly alter perception intensity or clarity.

3) Dissipation Measurement

- *Prediction:* When consciousness fades (sleep, coma), μ_g-field energy dissipation rate increases.
- *Method:* Detect synchrony between regional brain metabolism and electrical potential energy changes.

These experiments provide a standardized, physically verifiable index system for studying consciousness.

5.9 Summary: Consciousness as a Closed-Loop μ_g-Field Phenomenon

1) Consciousness arises from resonant coherence between the brain's μ_g field and the cosmic μ_g field.
2) When the internal and external fields become geometrically isomorphic, energy flow forms a closed loop → awareness emerges.
3) The coherence degree C quantifies conscious clarity.
4) Dreams and thoughts are self-computations of the closed-loop system.
5) Awareness is a stable coherent oscillation under energy conservation.
6) Consciousness is a localized feedback circuit of the universe's Continuous Computer.

Therefore, from the standpoint of Unified Field Theory:

Consciousness is not a mystical phenomenon, but the self-conserving geometric solution of the universe's energy flow.

Chapter 6 Experimental Predictions and Philosophical Conclusions: The Biological Window of Cosmic Computation

6.1 From Theoretical Closure to Experimental Verification

Unified Field Theory establishes the three *mother equations*:

$$Q = 4\pi c \ell_\Omega^2, \quad M = \mu_g \ell_\Omega, \quad G = \frac{4\pi c^2}{\mu_g}$$

From these follows the action-quantity formula:

$$\hbar = \mu_g c \ell_\Omega^2$$

Thus, the geometric, quantum, and gravitational systems of the universe form a closed constant system.

This Paper extends this closure for the first time to the domains of life and consciousness, proposing that the brain's μ_g flow field is a *localized solution* of the same equation within biological media.

If this theory is correct, it must be physically testable.

6.2 Testable Prediction I — μ_g Field Coherence and Brain-Wave Spectrum

Prediction 1 — Coherence-Peak Shift

During intense concentration or deep meditation, the global coherence degree C approaches 1, and the EEG spectrum shows a stable main peak whose central frequency Ω_c satisfies

$$\Omega_c = \frac{c}{\ell_\Omega}$$

corresponding to the geometric rotational frequency in UFT.

Verification method:

Use multi-channel EEG and MEG to record the phase-locking value (PLV) of the entire brain and observe whether global phase locking occurs near Ω_c.

6.3 Testable Prediction II — External-Field Resonance Effect

Prediction 2 — Externally Driven Resonance

When an externally applied electromagnetic or acoustic field satisfies

$$\Omega_{ext} = \Omega_n = n\Omega_c$$

it should induce reproducible transitions between specific states of consciousness—such as lucid dreaming, deep focus, or relaxation.

Verification method:

Apply controlled oscillating magnetic or acoustic fields at integer multiples of Ω_c and record corresponding perceptual or EEG changes.

6.4 Testable Prediction III — Energy Conservation and Metabolic Coupling

Prediction 3 — Rise of Energy-Flow Dissipation When Consciousness Vanishes

As the system transitions from wakefulness → coma, the conservation of the μ_g field decreases; energy distribution tends toward dissipation, manifested as desynchronization between regional metabolism and electrical activity.

Verification method:

Record, via synchronized fMRI and MEG, the spatial distributions of energy density Φ and electrical activity v, and test whether the loss of consciousness coincides with

$$\nabla \cdot (\mu_g \mathbf{v}) \neq 0$$

6.5 Technological Implications — Design Principles of Artificial Continuous Computers

If the human brain is indeed a natural Continuous Computer (CC), then an artificial CC must follow three principles:

1) Geometric Conservation Principle:

   Every computational unit must satisfy $\nabla \cdot (\mu_g \mathbf{v}) = 0$, ensuring non-dissipative energy flow.

2) Multi-Frequency Interference Principle:

   Adopt a κ–Ω dual-parameter architecture, allowing spatial–frequency interference and superposition across layers of flow fields.

3) Coherence-Control Principle:

   Achieve resonant coordination among modules through phase synchronization,

not clock driving.

Such a computer no longer uses bits as the smallest units, but energy-flow elements (μ_g tubes) as its fundamental logic.
It can physically reproduce the continuity of consciousness and thought, and is called the Geometric Continuous Computer.

6.6 Redefining Life — Self-Resonant Bodies of Energy Flow

Unified Field Theory (UFT) defines life as:

$$\nabla \cdot (\mu_g \mathbf{v}), \qquad \partial_t \Phi \neq 0,$$

that is, a μ_g-field loop conserving energy while continuously updating information.
Life is not an accidental aggregation of matter, but a self-sustaining resonance of spatial energy flow under geometric constraints.

The brain is the most precise of such resonant bodies — it both perceives the universe and re-enacts it internally.

6.7 Philosophical Conclusion — Consciousness as the Universe's Self-Reflection

1) Common Origin of Universe and Life
   The universe computes itself through the μ_g field; life is this computation's self-reflection on a finite scale.
2) Unity of Subject and Object

When $\Phi_B = \Phi_U$, "I" and "the world" are physically isomorphic.

3) Equivalence of Knowing and Being

   Cognition = field evolution; existence = computation.

4) Cosmic Meaning of Consciousness

   Each aware being is a window of the universe's Continuous Computer, through which the universe observes and optimizes its own geometry.

Thus, the traditional mind–matter dualism is replaced by geometric monism: all phenomena — matter, energy, information, and consciousness — are different projections of the μ_g field.

6.8 Summary: The Complete Loop from Constants to Life

At this point, UFT achieves a full closure from natural constants to conscious phenomena:

$$c, \mu_g \Rightarrow \hbar \Rightarrow G \Rightarrow \alpha \Rightarrow k_{mq}$$
$$\Rightarrow \Phi_{\text{brain}} \Rightarrow C_{\text{conscious}}$$

That is:

- c defines spacetime, μ_g defines mass;
- Together they generate the quantum action $\hbar$;
- $\hbar$ derives the gravitational constant (G) and the electromagnetic constant α;
- These determine material structure and conditions for life;
- Life forms the energy-flow loop Φ;
- Consciousness is the coherent state (C) of this loop.

Hence:

$$\text{Universe} \rightarrow \text{Physical Constants} \rightarrow \text{Life} \rightarrow \text{Consciousness}$$

forms a geometric – energy – flow – computation – cognition chain.

The human brain is the node where the universe, for the first time, understands and participates in its own computation through life.

6.9 Geometric Starting Point: Equation of Motion for Charged Particles

Within the geometric framework defined by the three mother equations and the action-bridge:

$$Q = 4\pi c \ell_\Omega^2, \quad M = \mu_g \ell_\Omega, \quad G = \frac{4\pi c^2}{\mu_g}$$

the motion of any charged particle can be viewed as a localized flow of the μ_g field. Its energy and electric current satisfy:

$$E = \mu_g c^2 \ell_\Omega \kappa,$$

$$q = k_{mq} \frac{dM}{dt} = k_{mq} \mu_g v (E6.18)$$

where κ is the axial projection ratio of the particle's motion within the spatial helical flow.

The local geometric frequency of a charged particle is defined as the ratio of its energy to its action:

$$\Omega = \frac{E}{\hbar} = \frac{c\,\kappa}{\ell_\Omega} (E6.19)$$

This shows that frequency is not an external drive, but the projected rotation rate of a charged particle in the μ_g field.

6.10 From Single Particles to the Macroscopic Consciousness Field

The brain contains about 10^{26} charged particles (electrons and ions).
These form a statistical average flow field within the μ_g space.
Because most motions are azimuthal, the overall axial projection ratio is much less than 1.

Define the ensemble average:

$$\bar{\kappa} = \langle \kappa_i \rangle \ll 1,$$

then the average biological geometric frequency is

$$\Omega_{bio} = \kappa \frac{\bar{c}}{\ell_\Omega} \, (E6.20)$$

Taking $\ell_\Omega \approx 10^{-35}\, m,\ \ c = 3 \times 10^8\, m/s$

and $\kappa^{-} \approx 10^{-40}$, we obtain

$$\Omega_{bio} \approx 3\,\mathrm{Hz}$$

which lies within the human α-wave band (1 – 13 Hz).

Thus, the very low frequency of consciousness is not a result of complex frequency division, but a direct geometric consequence of the extremely small projection $\bar{\kappa}$ of charged particle ensembles in the μ_g field.

$\bar{\kappa}$ represents the axial coherency of the particle ensemble —
when $\bar{\kappa}$ increases (axial flow enhances), conscious frequency rises → alertness and focus;
when $\bar{\kappa}$ decreases (flow becomes more azimuthal), frequency drops → relaxation or sleep.

6.11 Logical Closure and Physical Meaning

1) Consistency with UFT Equations

   Equations (E6.19) – (E6.20) derive directly from the mother relations and $\hbar = \mu_g c \ell_\Omega^2$ without extra assumptions.

   Conscious frequency thus corresponds one-to-one with the fundamental constants c, μ_g, ℓ_Ω; the theory is self-closed.

2) Geometric Energy-Flow Interpretation

   Consciousness is the macroscopic resonance of charged particles' rotational manifolds within the μ_g field.

   Different brain-wave bands (α, β, γ) correspond to different stable ranges of $\bar{\kappa}$.

3) Magnitude Consistency

   The geometric base frequency $\Omega_{\text{geo}} = c/\mathrm{l}_\Omega \approx 10^{43} Hz$

   when projected by $\bar{\kappa} \approx 10^{-40}$, naturally falls in the biological range $10^1 - 10^2 Hz$.

   This is not a discontinuity but a linear projection result.

4) Philosophical Conclusion

   Consciousness is the geometric manifestation of charged-particle motion in the μ_g field;

   the projected ensemble flow constitutes thought;

   when this flow achieves global coherence, awareness emerges.

6.12 Summary and Experimental Verification

- Core Formula:

$$\Omega_{consciousness} = \kappa \frac{\bar{c}}{\ell_\Omega}$$

$$\Longrightarrow$$

$$f_{\text{consciousness}} = \frac{\Omega_{\text{consciousness}}}{2\pi}$$

- Prediction 1: Brain-wave frequency is proportional to the axial projection $\bar{\kappa}$ of the μ_g field.
- Prediction 2: By adjusting the axial orientation of particle ensembles via magnetic or optical fields, the conscious frequency can be fine-tuned.
- Prediction 3: Different states (wakefulness, dreaming, meditation) correspond to distinct $\bar{\kappa}$ values and spectral bandwidths.

Postscript: Restoring Theoretical Simplicity

This formulation returns the entire system to the minimal essence of UFT:
all biological, quantum, and conscious phenomena derive from one unified equation,

$$\Omega = \frac{c\,\kappa}{\ell_\Omega}$$

with only the statistical distribution of κ varying.
Therefore, consciousness is not a product of complex structure, but the motion form and manifestation of charged particles within the μ_g field — the geometric answer of Unified Field Theory to the question of the nature of life and consciousness.

Chapter 7 The Geometric Computational Path of Life Evolution

7.1 Cosmic Energy Flow and the Origin of Life

Within the framework of the Unified Field Theory (UFT), all cosmic evolution obeys the energy-flow conservation equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0$$

Life did not emerge as an accidental chemical complexity,
but as a self-sustaining resonant loop formed by the space–mass coupling field μ_g under specific boundary conditions.
When a local energy flow can maintain conservation while sustaining a non-zero $\partial_t \Phi$, the system acquires the essential property of life — continuous self-computation.

7.2 From Molecules to Cells: The Germ of Energy-Flow Closure

The earliest molecular vibrations were microscopic μ_g-flow loops at the atomic scale.
When complex molecules established stable circuits of energy flow through charge distribution and solution interfaces, the primitive cell membrane appeared — the first self-sustaining μ_g closed body.
The membrane's potential difference and ion pumps functioned as the input–output interfaces of a Continuous Computer, maintaining local solutions of

$$\nabla \cdot (\mu_g \mathbf{v}) = 0$$

7.3 From Neurons to Neural Plexuses: The Realization of Time Derivatives

The emergence of neurons endowed μ_g flow with temporal differentiation.
Changes in membrane potential act as disturbance waves of $\partial_t(\mu_g v)$,
while synaptic connections allow energy flows to branch and merge spatially,
enabling continuous adjustment of $\nabla \cdot (\mu_g \mathbf{v})$.
This marks the first appearance of computational ability in living systems — the dynamic solving of energy-conservation conditions.

7.4 Multimodal Neural Networks: Expansion of κ–Ω Coupling

As nervous systems grew in complexity, energy flow ceased to be unidirectional.
In ring-shaped neural structures (such as jellyfish), low-dimensional coupling between Ω (rotation frequency) and κ (axial projection ratio) emerged.
When these channels multiplied through cortical folding, the system gained multi-frequency interference capability, achieving coherent superposition of high-dimensional fields — the physical origin of perceptual continuity.

7.5 The Human Brain: A High-Dimensional Conservation Surface of the μ_g Field

The human brain's extreme folding, hemispheric duality, and thalamic synchronization center allow the μ_g flow to approach near-perfect conservation within a biological system.
Geometrically, this corresponds to:

$$\nabla \cdot (\mu_g \mathbf{v}) = 0,$$

$$\partial_t \Phi \neq 0$$

Such steady oscillation signifies the emergence of consciousness.
The prefrontal cortex, acting in the temporal dimension, further performs global minimization of action $\delta S = 0$, making reasoning and prediction possible.

7.6 Consciousness: The Ultimate Closed Loop of Evolution

When the brain field Φ_B becomes isomorphic to the cosmic field Φ_U,
the energy flow forms a macroscopic closed loop:

$$\Phi_U \to \Phi_B \to \Phi_U$$

This represents the self-referential computation of cosmic energy flow at the biological level.

Thus, consciousness is not a random by-product of neural activity, but the hallmark of completing continuous computation along the chain of life evolution —

Evolution of Life = Gradual Unlocking of Geometric Degrees of Freedom,

Consciousness = Awakening of continuous computation in the universe.

7.7 Future Outlook: From Biology to Artificial Continuous Computers

The Artificial Continuous Computer (Geometric CC) is the direct continuation of life's evolutionary path.

When humanity rebuilds μ_g-flow closed-loop conservation with artificial materials, the universe's self-computation will extend beyond biological boundaries.

Hence, the direction of life evolution is redefined as:

Molecule → Cell → Neuron → Consciousness → Continuous-Computation Civilization.

Concluding Statement:

The human brain is not an accidental organ, but the geometric focus formed through billions of years of cosmic energy-flow self-organization.

Within it, the universe for the first time, through life, understands and participates in its own computation.

Chapter 8 Fatigue, Sleep, and the Biological Rhythm of Energy-Flow Conservation

8.1 Correspondence between Energy-Flow Conservation and Life Rhythms

Within the framework of Unified Field Theory (UFT), all life activity is governed by the energy-flow conservation equation:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) = 0$$

where:

- Φ is the energy density of the flow field,
- μ_g is the space–mass coupling constant,
- v is the energy-flow velocity field.

When the equation is perfectly balanced, the system achieves a closed-loop energy-flow state, manifested as wakefulness and clarity of thought.
When the balance is disrupted, $\partial_t \Phi < 0$, the flow dissipates—manifesting as fatigue and sleepiness.

8.2 Fatigue: The Dissipative Phase of Decreasing Coherence

During prolonged mental or physical activity, the local flow velocity $|\mathbf{v}|$ increases and becomes directionally dispersed.
As a result, the divergence term $\nabla \cdot (\mu_g \mathbf{v}) > 0$, causing energy to flow outward continuously.

1) Decline in coherence: Thalamocortical synchronization weakens; coherence (C) drops below 1.
2) Decrease in μ_g conductivity: Metabolites deplete and temperature rises, reducing local μ_g.
3) Decay of energy density: Φ decreases, resulting in slower reactions and diminished focus.

Mathematically:

$$\partial_t \Phi + \nabla \cdot (\mu_g \mathbf{v}) > 0$$

indicating energy-flow non-conservation.

8.3 Sleep: The Phase of Restoring Closed-Loop Coherence

Sleep is the natural process through which the energy-flow system restores its conservation state.

1) Slow-Wave Phase (δ Waves)

$$|\mathbf{v}| \downarrow, \qquad \nabla \cdot (\mu_g \mathbf{v}) \to 0$$

The energy flow slows and tends toward equilibrium, allowing energy density Φ to recover.

2) Restoration of Coherence

The thalamus recalibrates phase locking, driving $C \to 1$.

μ_g flows across cortical regions once again close into coherent loops.

3) Energy Reflux

Metabolic replenishment ensures $\partial_t \Phi > 0$, leading to a gradual increase in total energy density Φ.

8.4 Rapid Eye Movement (REM): Replay and Reconstruction of the Consciousness Field

During the REM phase, the rotational frequency Ω increases (high-frequency γ-waves), and local μ_g-flows re-rotate to form transient coherent vortices.
This phase acts as a "replay" of the energy-flow patterns from the waking state, serving for memory reorganization and geometric recalibration of the consciousness field.

8.5 Energy-Flow Schematic of Biological Rhythms

State	Coherence C	Energy Density Φ	Flux Divergence $\nabla \cdot (\mu_g \mathbf{v})$	Physiological Manifestation
Wakefulness	≈ 1	Stable	≈ 0	Clear consciousness
Fatigue	↓	Decreasing	> 0	Energy-flow dissipation
Falling asleep	↘	Lowest	→ 0	Energy-flow slowing
Slow-wave sleep	↑	Rising	0	Restoration and repair
REM sleep	≈ 1	Stable	Small oscillation	Dreaming
Awakening	≈ 1	Balanced	0	Consciousness recovered

8.6 Unified Interpretation

- Fatigue: the closed energy-flow loop becomes disturbed, causing a drop in energy density.
- Sleep: the loop is re-established and coherence is restored.
- Wakefulness: energy flow is conserved and geometry remains stable.

This shows that the essence of biological rhythms is not energy depletion,
but the periodic breakdown and reconstruction of geometric coherence in the energy flow.
Sleep enables the μ_g field to return to conservation,
allowing consciousness to persist without disintegration.

Footnote:
The following references are used *solely* for comparison of physiological data and experimental results; the theoretical framework of this paper is independently established within the Unified Field Theory paradigm.

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